OPERATIONAL CHOW COHOMOLOGY OF A SCHEME OVER A BASE OF DIMENSION ONE IS CHOW HOMOLOGY IN COMPLEMENTARY DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FC5 (sources/stacks-chow/pinned-original-source.tex line 14734).

Allowed standard prerequisites (editorial).

- Cycles and their presentation as sums of integral closed subschemes. Same chapter, Definition labelled definition-cycles, cited at proof lines 14752-14754. Provides the decomposition alpha = sum n_i [X_i] with g : coprod X_i -> X proper.
- Flat pullback, Gysin maps and proper pushforward as bivariant classes. Same chapter, Lemma labelled lemma-push-proper-bivariant, cited at proof lines 14776-14778, with the flat pullback and Gysin classes used at proof lines 14758-14768. Builds xi_i in the two cases and shows that g_* o xi is a bivariant class.
- Testing bivariant classes on integral closed subschemes. Same chapter, Lemma labelled lemma-bivariant-zero, cited at proof lines 14792-14794. Reduces the inverse identity to the case of an integral X'.
- The two shapes of an integral scheme over a one dimensional base. Same chapter, the construction of ord and the cycle theory for the situation, invoked at proof lines 14796-14802. Gives [X'] = (p')^*[S]_1 when p' is flat of relative dimension dim_delta(X') - 1, and [X'] = (p')^*(s -> S)^*[S]_1 otherwise.

Source reference. Original definition 02QR: sources/stacks-chow/pinned-original-source.tex, source lines 1535–1535. Same-chapter pointer into the author environment definition at source line 1535; invoked at proof line 14757 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 14781 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FC4: sources/stacks-chow/pinned-original-source.tex, source lines 14658–14658. Same-chapter pointer into the author environment lemma at source line 14658; invoked at proof line 14784 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 14797 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as above. Let X be a scheme locally of finite type over S. Then we have a canonical identification*

$$A^p(X \to S) = \mathrm{CH}_{1-p}(X)$$

for all $p \in \mathbf{Z}$.

Proof. Consider the element $[S]_1 \in \mathrm{CH}_1(S)$. We get a map $A^p(X \to S) \to \mathrm{CH}_{1-p}(X)$ by sending c to $c \cap [S]_1$.

Conversely, suppose we have $\alpha \in \mathrm{CH}_{1-p}(X)$. Then we can define $c_\alpha \in A^p(X \to S)$ as follows: given $X' \to S$ and $\alpha' \in \mathrm{CH}_n(X')$ we let

$$c_\alpha \cap \alpha' = \alpha \times \alpha'$$

in $\mathrm{CH}_{n-p}(X \times_S X')$. To show that this is a bivariant class we write $\alpha = \sum_{i \in I} n_i [X_i]$ as in Definition 02QR. In particular the morphism

$$g : \coprod\nolimits_{i \in I} X_i \longrightarrow X$$

is proper. Pick $i \in I$. If X_i dominates an irreducible component of S, then the structure morphism $p_i : X_i \to S$ is flat and we have $\xi_i = p_i^* \in A^p(X_i \to S)$. On the other hand, if p_i factors as $p_i' : X_i \to s_i$ followed by the inclusion $s_i \to S$ of a closed point, then we have $\xi_i = (p_i')^* \circ c_i \in A^p(X_i \to S)$ where $c_i \in A^1(s_i \to S)$ is the Gysin homomorphism and $(p_i')^*$ is flat pullback. Observe that

$$A^p(\coprod\nolimits_{i \in I} X_i \to S) = \prod\nolimits_{i \in I} A^p(X_i \to S)$$

Thus we have

$$\xi = \sum n_i \xi_i \in A^p(\coprod\nolimits_{i \in I} X_i \to S)$$

Finally, since g is proper we have a bivariant class

$$g_* \circ \xi \in A^p(X \to S)$$

by Lemma 0EPK. The reader easily verifies that c_α is equal to this class (please compare with the proof of Lemma 0FC4) and hence is itself a bivariant class.

To finish the proof we have to show that the two constructions are mutually inverse. Since $c_\alpha \cap [S]_1 = \alpha$ this is clear for one of the two directions. For the other, let $c \in A^p(X \to S)$ and set $\alpha = c \cap [S]_1$. It suffices to prove that

$$c \cap [X'] = c_\alpha \cap [X']$$

when X' is an integral scheme locally of finite type over S, see Lemma 02UC. However, either $p' : X' \to S$ is flat of relative dimension $\dim_\delta(X') - 1$ and hence $[X'] = (p')^*[S]_1$ or $X' \to S$ factors as $X' \to s \to S$ and hence $[X'] = (p')^*(s \to S)^*[S]_1$. Thus the fact that the bivariant classes c and c_α agree on $[S]_1$ implies they

agree when capped against $[X']$ (since bivariant classes commute with flat pullback and Gysin maps) and the proof is complete. $\square$